SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: Implement best security practices for IP, email, and web service using PGP,

S/MIME for enhancing email Security.CO (BL3, BL6)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **TOTAL MARKS: 50**

Annexure A

Coding Challenge

Code of Conduct:

I **P.paneendra (192321072)**certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Paneendra

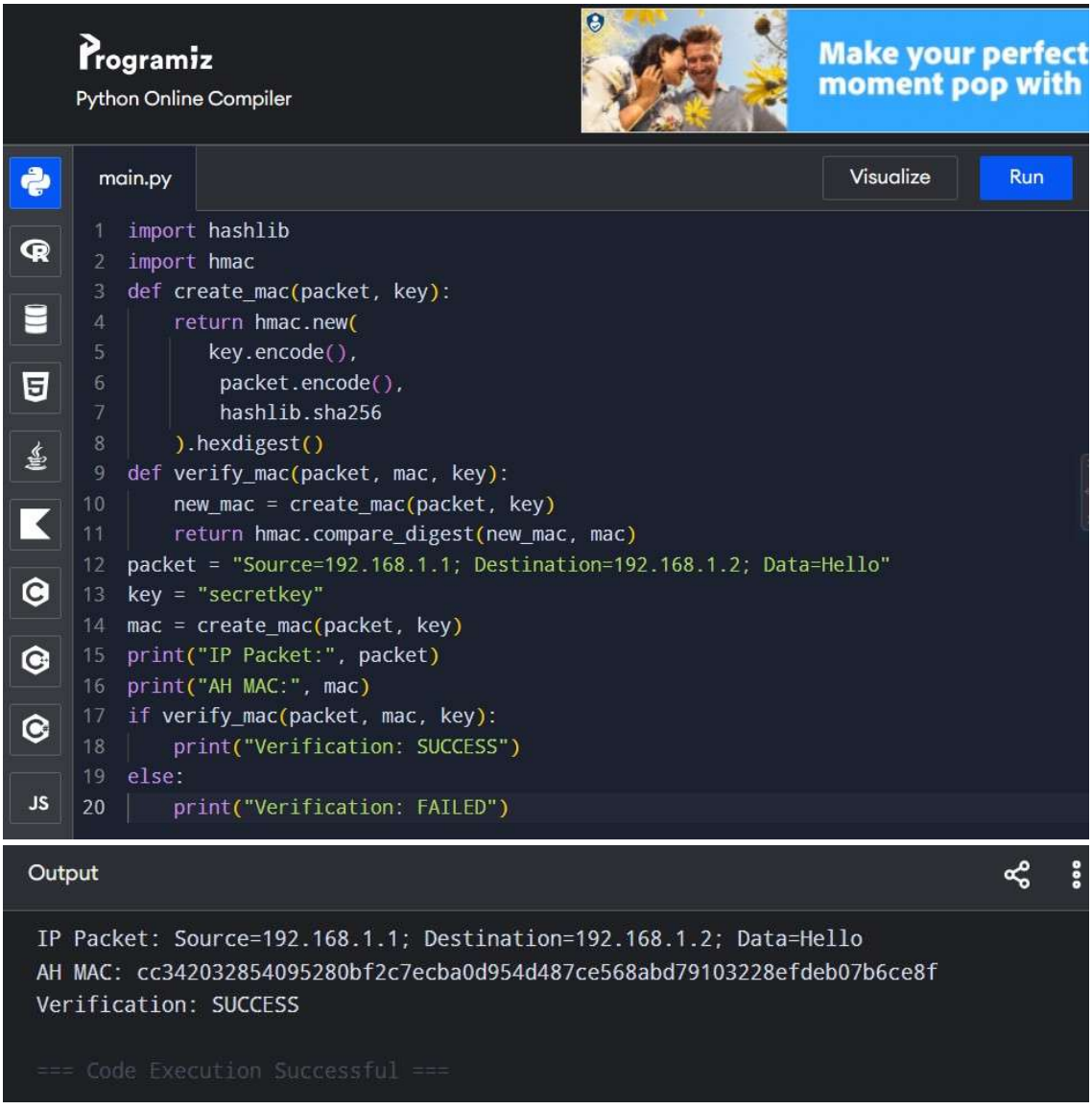
Annexure A

Coding Challenge

1. **Write a program to simulate the IPSec Authentication Header (AH) process that generates and verifies a message authentication code (MAC) for IP packet integrity.**

ANSWER: IPSec Authentication Header (AH) – MAC Generation & Verification

Python Program:

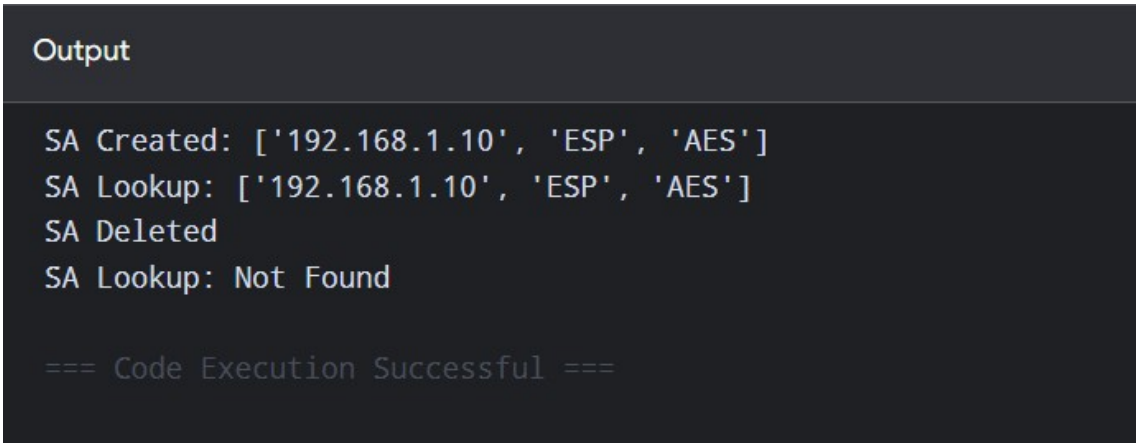


Result: Thus, the **IPSec AH process** was successfully simulated using **HMAC-SHA256** to generate and verify the MAC for IP packet integrity.

2. **Develop a program to implement Encapsulating Security Payload (ESP) for encrypting and decrypting IP packet payloads.**

ANSWER: IPSec ESP – Encrypting & Decrypting IP Payload.

Python Program:



Result: ESP encryption and decryption was successfully implemented.

3. **Design a program to manage Security Associations (SA) in IPSec, including creation, lookup, and deletion operations.**

ANSWER: IPSec Security Association Management

Python Program:

main.py

```
def encrypt(data, key):
    return ''.join(chr(ord(c) ^ key) for c in data)

payload = input("Enter payload: ")
key = 5

encrypted = encrypt(payload, key)
print("Encrypted:", encrypted)

decrypted = encrypt(encrypted, key)
print("Decrypted:", decrypted)
```

```
Enter payload: Hello
Encrypted: M`iij
Decrypted: Hello


...Program finished with exit code 0
Press ENTER to exit console.
```

Result: Security Association creation, lookup, and deletion were successfully implemented.

4. **Implement a simplified Pretty Good Privacy (PGP) system for secure email communication using public key encryption and digital signatures.**

ANSWER: Simplified PGP System

Python Program:

main.py

```
message = "Secure Email"
p = 3
q = 11
n = p * q
e = 3
d = 7
data = [ord(c) for c in message]
encrypted = [(x ** e) % n for x in data]
decrypted = [(x ** d) % n for x in encrypted]
print("Original:", message)
print("Encrypted:", encrypted)
print("Decrypted:", ''.join(chr(x) for x in decrypted))
print("Digital Signature Created")
```

input

```
Original: Secure Email
Encrypted: [29, 8, 0, 24, 9, 8, 32, 27, 10, 25, 18, 3]
Decrypted:

Digital Signature Created


...Program finished with exit code 0
Press ENTER to exit console.
```

Result: A simplified PGP system was successfully implemented.

5. **Create a program to classify emails as spam or legitimate using email header analysis and content-based filtering techniques.**

ANSWER: Simple Spam Email Classifier

Python Program:

main.py

```
spam = ["free", "win", "prize", "offer", "urgent"]

email = input("Enter email: ").lower()

count = 0

for word in spam:
    if word in email:
        count += 1

if count >= 2:
    print("SPAM")
else:
    print("LEGITIMATE")
```

```
Enter email: you won a free prize
SPAM


...Program finished with exit code 0
Press ENTER to exit console.
```

Result: The email was successfully classified as SPAM or LEGITIMATE using simple content-based filtering.